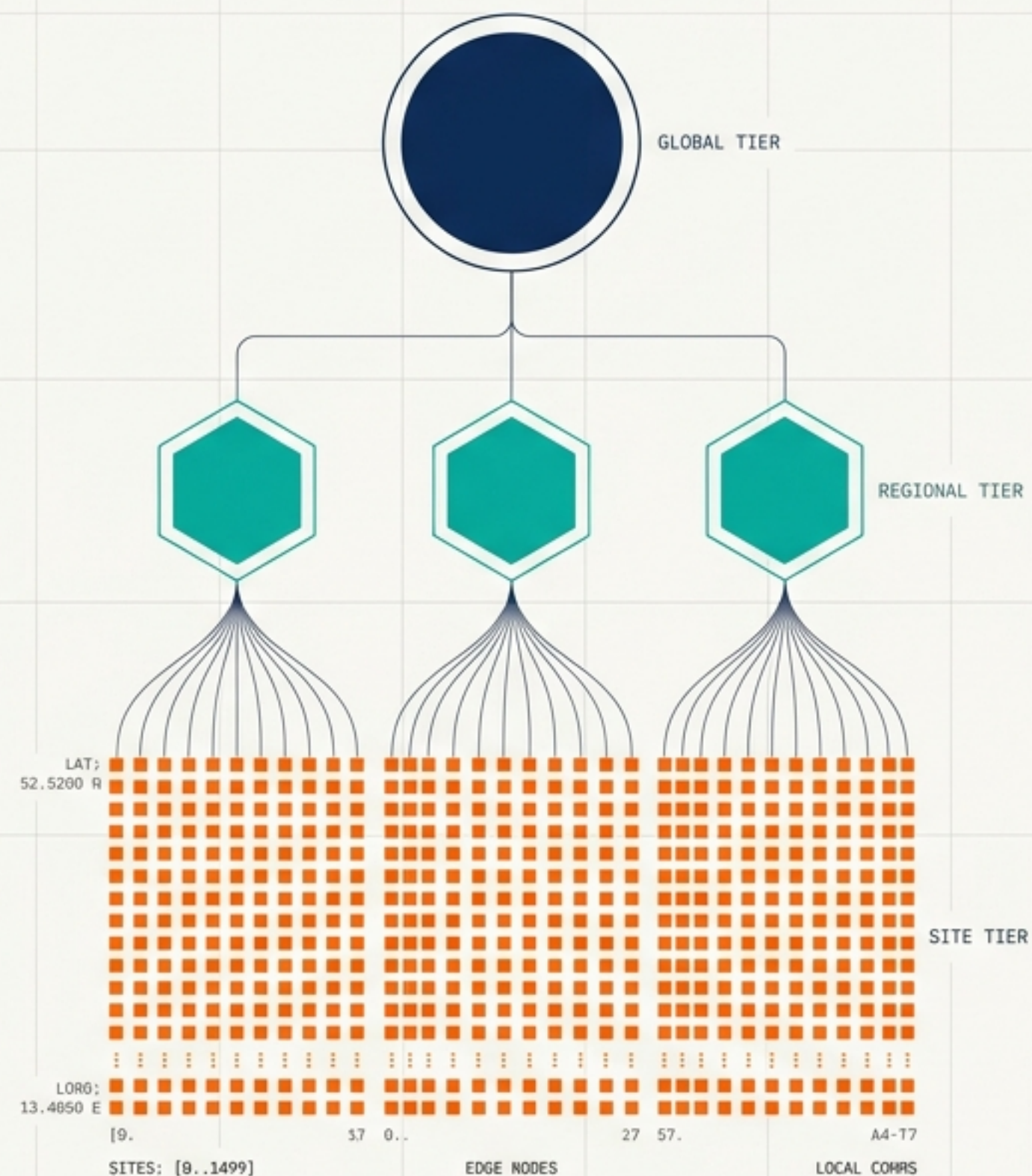


AMQ Fleet Control Plane

Pre-Test Architecture Design · Rev 2



Draft for demo planning · 15 Sep 2026 · Target Scale: 1,500 Sites

X-14.25
Y+89.80
Z-62.10

Global (Intent)

Multicluster Global
Hub operator

PostgreSQL / Grafana
(inventory/compliance only)

Global OpenShift GitOps
(Push ApplicationSet to 3 hubs)

Global Thanos Query

Regional (Work)

ACM
(MultiClusterHub)

Regional OpenShift GitOps
(Pull ApplicationSet for sites)

MultiClusterObservability
(Thanos receive, GCS bucket,
Alertmanager)

Site (Run)

Klusterlet

Site OpenShift GitOps

AMQ Broker Operator

AMQ Broker (1-2 pods)

User Workload Monitoring

X-14.25
Y+89.80
Z-82.10

Rule of execution: Sites never talk to the global hub.

X-14
Y+89
Z-82

4.25
9.80
2.10

X-14.25
Y-89.89
Z-82.10

GLOBAL HUB

Global Kafka
bootstrap

Global Argo CD

Push to 3 regional APIs

REGIONAL HUBS

Regional hub

Regional
Thanos

Regional
hub agents

policies/status

metrics

SITES

Site
klusterlet

Site
Prometheus

Site
Argo

Git

X-14.25
Y-89.89
Z-82.10

GCP ROUTING CONSTRAINTS

- Regional Thanos and Kafka routes are strictly GCP-to-GCP, keeping traffic on private networks.
- Sites require outbound HTTPS only.
- Zero inbound connections are ever opened to a site.

X-14.25
Y-89.89
Z-82.10

X-14.25
Y-89.89
Z-82.10

Ownership Matrix: Platform vs. Messaging

Platform Team / Slow Changes

- Authors at Global via ACM Policy.
- Controls: Hub installs, Operator version pins, namespaces, network policies.
- Rationale: Requires fleet-wide compliance reporting.

Messaging Team / Fast Changes

- Authors via Git Pull Requests.
- Controls: Broker definitions, addresses, queues, security, service monitors.
- Rationale: Requires diffs, code review, and instant rollback.

CODE SNIPPET

```
spec.brokerProperties:  
  - addressConfigurations.'orders.inbound'.routingTypes=ANYCAST  
  - addressConfigurations.'events.telemetry'.routingTypes=MULTICAST
```

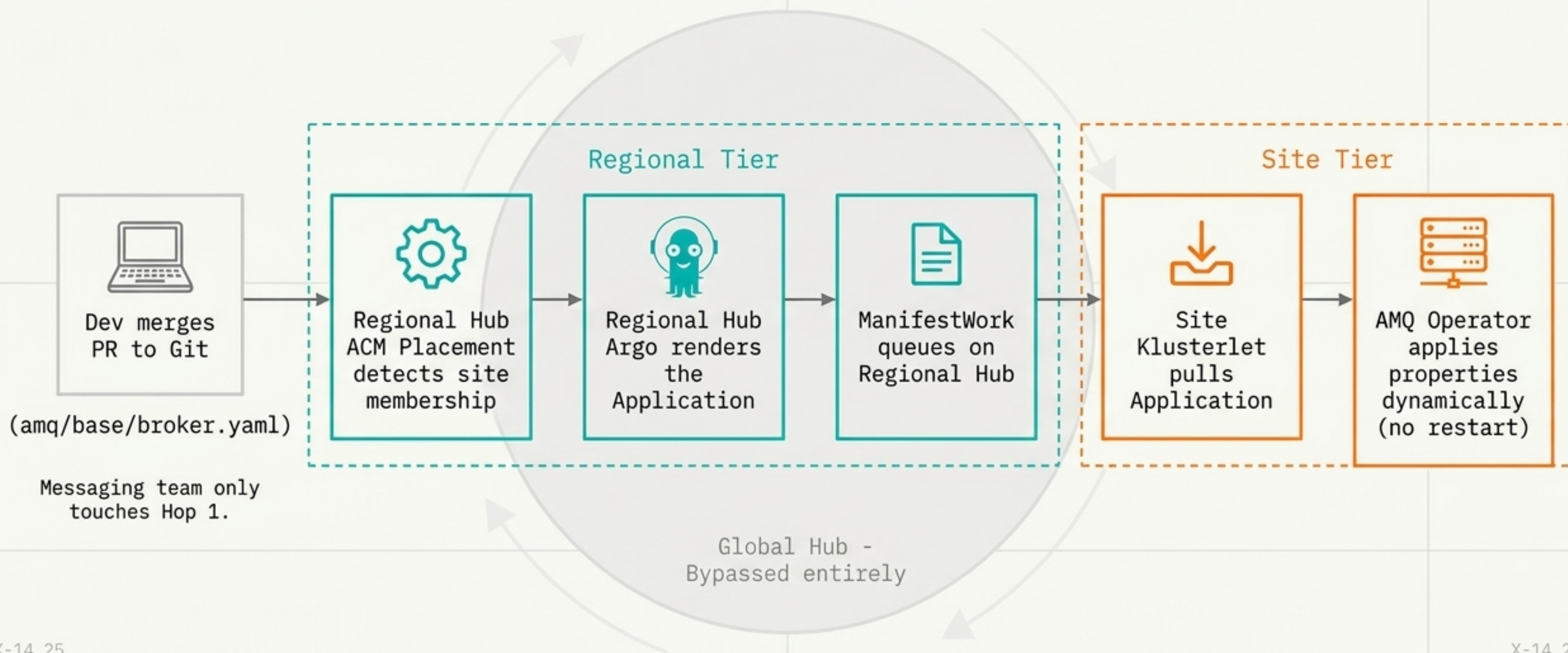
Legacy address CRD
is deprecated.

GitOps Strategy: Push vs. Pull

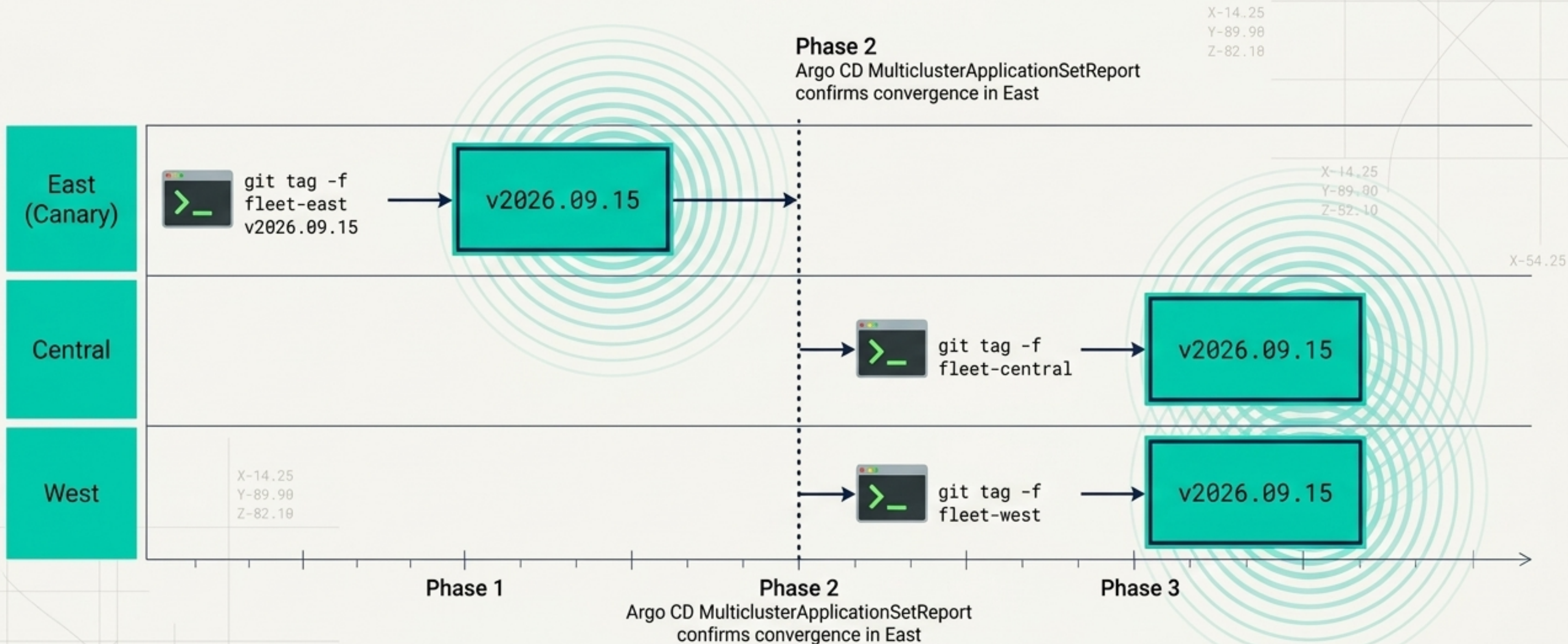
Dimension	Push (Global -> Regions)	Pull (Regions -> Sites)
Network Reachability	Hub must reach target API	Site reaches Git (Outbound only)
Hub Load	Fine for 3 hubs	Distributed rendering/syncing load across 1,500 site CPUs, preventing regional hub memory exhaustion
Target Outages	Manifests queue on the hub	Site independently reconciles from last known Git state upon reconnection

Pull is the only mathematical choice for the 1,500-site tier. OpenShift GitOps footprint (~1 GiB memory) is an acceptable trade-off for distributed sync.

The GitOps Pipeline: Managing Topics via Code



Deployment Strategy: Blast-Radius-Aware Rollout



A bad change is simply a tag moved back.

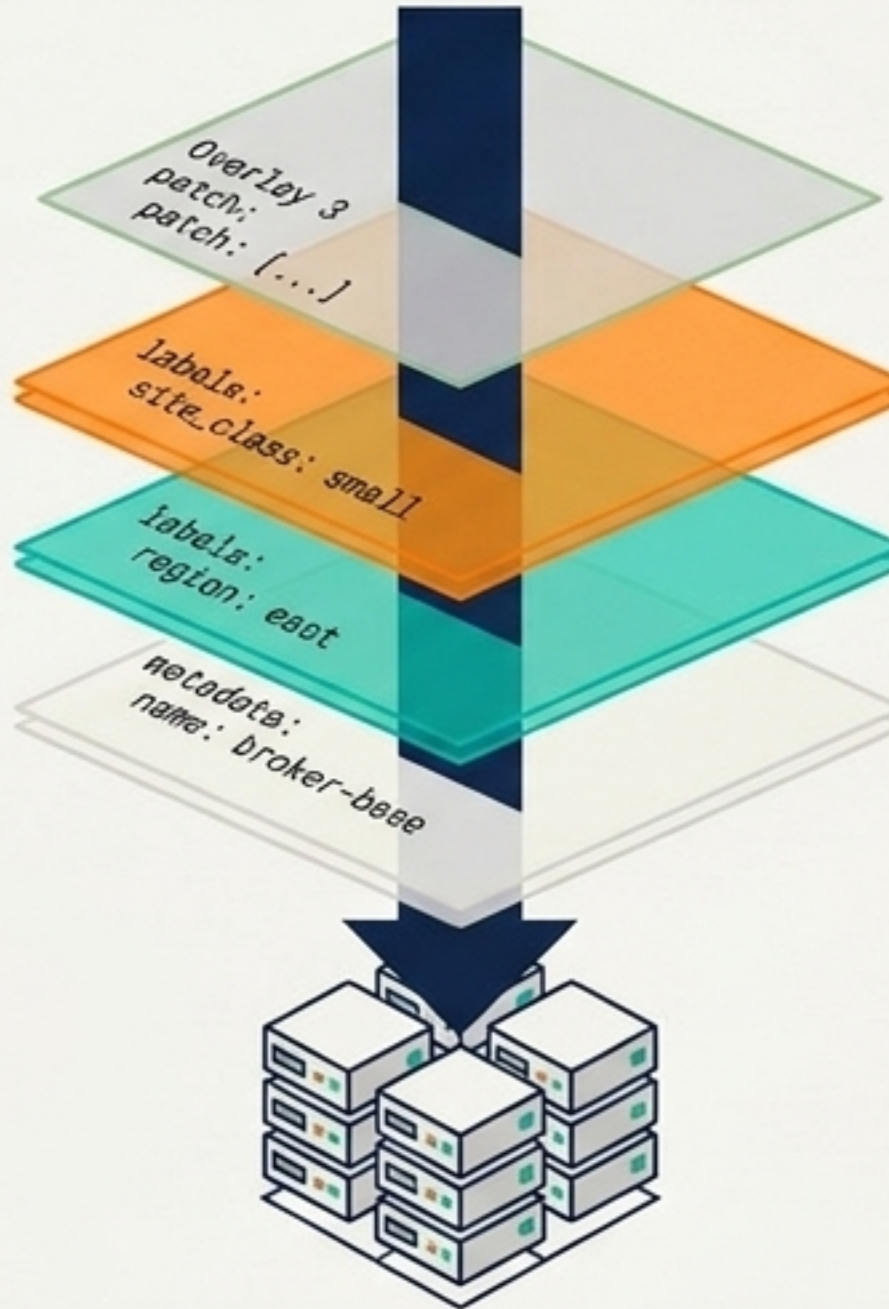
Broker Provisioning: The Configuration Patchwork

Overlay 3 (Feature Label):
appends addresses-returns JSON patch

Overlay 1 (Region Label):
targetRevision pinned to fleet-east

Overlay 2 (Site Class Label):
size: 1 (Small) OR
broker-ha-pair (Large)

Base Layer: broker.yaml
(Common properties)



Site variation is expressed as data in ACM labels, not forks in Git. A site upgrade from small to HA-pair is a single label change, requiring zero commits.

Observability Architecture: The Metric Funnel

X-14.25
Y-89.98
Z-82.18

Y-15.25

Site Prometheus:
Retains full-resolution data locally.
Scrapes via ServiceMonitor.

Insight

A single PromQL query from the **Global Grafana** fans out to three regions and three regions and merges instantly, stamping the region as a **label**.

The Allowlist Filter:
uw1_metrics_list.yaml allows only
artemis_* series to cross the
WAN to the Regional GCS Bucket.

Global Thanos Query:
Queries the 3 regional gRPC
services on demand.
Stores absolutely nothing.

X-14.25
Y-89.98
Z-82.18

X-14.25
Y-89.98
Z-82.18

Z-24.48

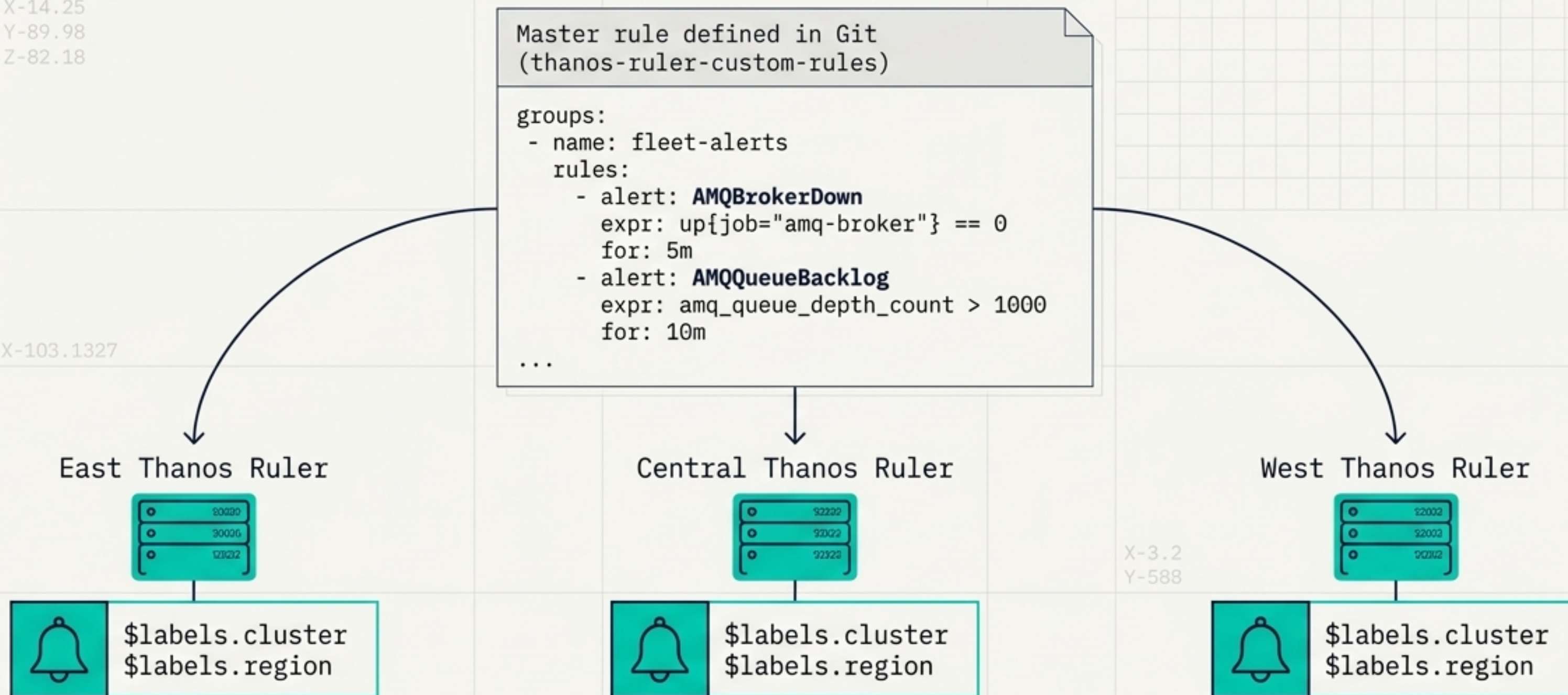
Alerting Governance: Fleet Alerts

X-14.25
Y-89.98
Z-82.18

X-14.25
Y-89.98
Z-82.18

X-103.1327

X-3.2
Y-588



One rule definition, three execution engines. Alerting survives a total loss of the Global Hub or global connectivity.

Failure Domains: The Decoupled Fleet

	Sites & Brokers	Config Delivery	Metrics & Alerts
Global Hub Down	✓ Unaffected	Topic changes flow. Fleet policies wait.	Regional dashboards work. Fleet-wide Grafana lost.
One Regional Hub Down	✓ Unaffected	The Aha Moment: Control Plane is entirely decoupled from the Data Plane. Brokers run unharmed.	
Git Down	✓ Unaffected	Site Argo CDs hold last known state.	
One Site Down	✓ Unaffected		

X-14
Y-89
Z-82

X-14.25
Y-89.98
Z-82.18

Scale & Capacity: Managing 1,500

X-14.25
Y-89.98
Z-82.18

Table 1: Metric Volume

Assumptions	1 broker + 20 addresses per site = 175 series per site.
Fleet Scale	1,500 sites = 262,500 total series.
Load	At 300s intervals = 875 samples per second fleet-wide.

X-14.25
Y-89.98
Z-82.18

Sizing Constraints

A regional hub handling ~500 sites is well within tested limits (3,500 single-node clusters on one hub).

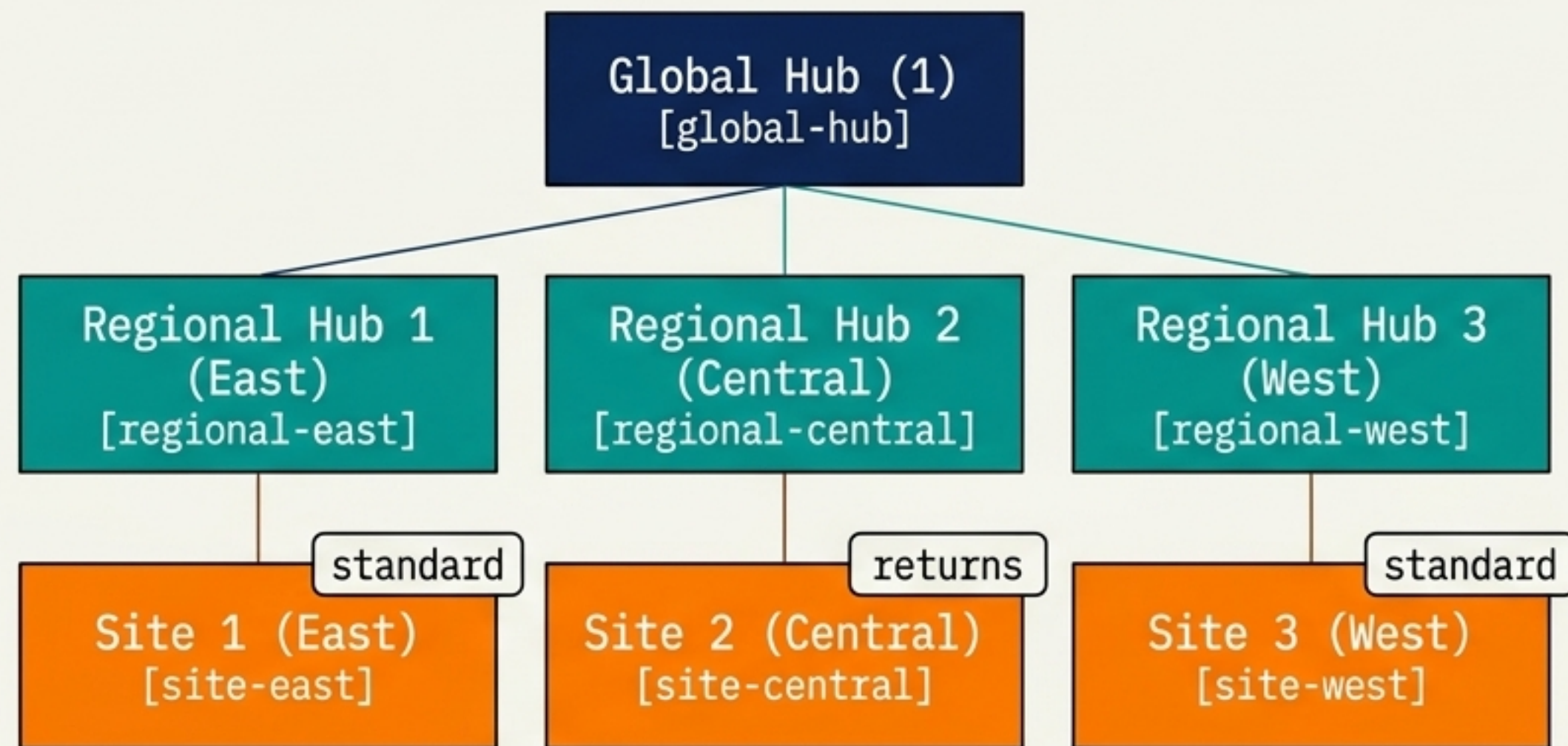
GCP Architecture Details

Regional GCS buckets keep Thanos traffic local. Metric ingress into GCP is strictly free.

The Pre-Test Plan

X-14.25
Y-89.98
Z-82.18

7-Cluster Topology



The Script

1. PR merged for `orders.returns`. (Nothing changes).
2. Move `fleet-east` tag. (East site syncs).
3. Validate `artemis_message_count` in East Grafana.
4. Scale Global Hub to zero. Move Central tag. (Proves messaging deployment survives management outage).

Pending Customer Decisions

- ✓ GCP OpenShift flavor
- ✓ HA pair requirements
- ✓ Metric retention policies